INSTITUTO DE BIOMEDICINA DE VALENCIA CSIC

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# Detección genómica de la resistencia a antibióticos

## Práctica

## Máster en Biotecnología Biomédica

## Introduction

Growing rates of antimicrobial resistance make antibiotic susceptibility testing (AST) increasingly needed to ensure the right antibiotics are prescribed for patients with bacterial infections. Determining antibiotic susceptibility is preferred over empiric therapy, wherein typically broad-spectrum drugs are used without a definitive confirmation of the infectious agent and which antibiotics infectious bacteria are resistant to. Data collected on antibiograms (strains' full susceptibility pattern) can also be used for surveillance purposes and, in turn, inform empiric therapy. AST is routinely performed using culture-based techniques in clinical diagnostic laboratories, frequently disk diffusion, broth microdilution and gradient diffusion (i.e. E-test). As antibiotic resistance is genetically encoded, i.e. mediated by acquisition of new genes, gene copy number and mutations in regulatory and coding regions, molecular tests have been developed to target the detection of such genetic markers.[4] In the last decade, **whole-genome sequencing** has emerged as an alternative technology to both culture and targeted molecular tests for the diagnosis of AMR as it can, in principle, detect all AMR genetic determinants and predict resistance to all antibiotics in a single experiment. The **accuracy of genotypic predictions** depends on the availability of: (1) accurate catalogues of AMR genetic determinants, (2) large collections of whole-genome sequenced strains with AST measurements to assess the diagnostic accuracy of such catalogues, and (3) automated genome analysis and interpretation tools.

**Mutational (chromosomal) resistance** is the main driver of acquired resistance in certain bacterial species, such as *Mycobacterium tuberculosis* and *Helicobacter pylori*, or for particular antibiotics, especially to synthetic agents such as fluoroquinolones and oxazolidinones. Resistance mutations are vertically transmitted, i.e. via clonal reproduction of bacteria, or can be transmitted horizontally via homologous recombination between different strains. **Gene-mediated resistance** is the main driver of acquired resistance in certain bacterial species, particularly gram-negatives. Resistance genes can be horizontally transmitted (via vectors such as plasmids) and also vertically transmitted (particularly stable if integrated into the chromosome). In some bacterial species, chromosomal and gene-mediated resistance are equally common (e.g. *Staphylococcus aureus*). Resistance to the same antibiotic can be conferred by both mutations and acquired genes (e.g. fusidic acid in *Staphylococcus aureus*, colistin resistance in *Escherichia coli*).

Over the years, a number of global studies have identified the genes and mutations that confer resistance to particular antibiotics. There are several databases such as the **Comprehensive Antimicrobial Resistance Database (CARD)** (https://card.mcmaster.ca/), **ResFinder** (https://cge.cbs.dtu.dk/services/ResFinder/), **AMRFinder** (https://www.ncbi.nlm.nih.gov/pathogens/antimicrobial-resistance/AMRFinder/) **or Pathogenwatch** (https://pathogen.watch/) that contain information about the genes and mutations that confer resistance. The use of these databases and tools depends on the species and mechanisms of resistance one is interested in.

# Case study 1: Identifying linezolid-resistance point mutations from *Enterococcus faecium* genomes

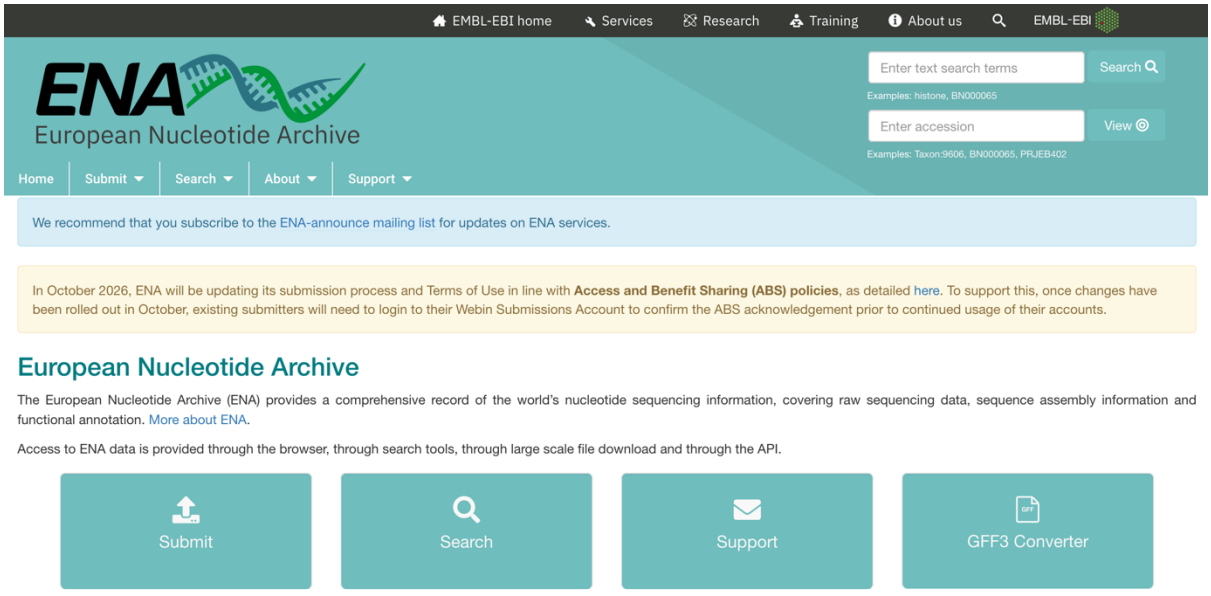
Linezolid, an oxazolidinone antibiotic that inhibits protein synthesis by binding to the 50S subunit of the 23S rRNA, is considered a last-resort treatment option for vancomycin-resistant Enterococci (VRE). However, linezolid resistance has been reported in *Enterococcus faecium*, with the most prevalent resistance mechanisms involving mutations within the V domain of the 23S rRNA. *E. faecium* generally carries six 23S rRNA gene copies, and mutations affecting the linezolid-binding region—particularly G2576T—can confer resistance, with the proportion of mutated copies influencing its level. In this exercise, we will analyse the genomes of four linezolid resistant VRE strains that originated from a recent study. We will evaluate the ability of different bioinformatic tools to detect genetic markers of linezolid resistance from Illumina sequence data, from both raw sequence reads and genome assemblies. The table below includes the accessions, isolate Ids, and linezolid MIC of the VRE genomes we will analyse:

**Table 1**: linezolid resistant *E. faecium* strains to analyse in this exercise.

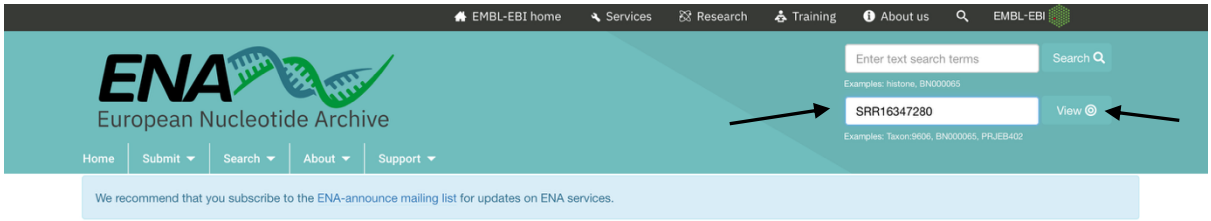
| Illumina run accession | Isolate ID | Biosample accession | Linezolid MIC | Linezolid phenotypic resistance | Resistance Genotype |
|---|---|---|---|---|---|
| SRR16347280 | Efm-136 | SAMN22135218 | 1 | S | |
| SRR16347263 | Efm-151 | SAMN22135233 | 16 | R | |
| SRR16347013 | Efm-377 | SAMN22135459 | 8 | R | |
| SRR16346985 | Efm-59 | SAMN22135141 | NA | R | |

## 1.1 Downloading raw sequence reads from the European Nucleotide Archive

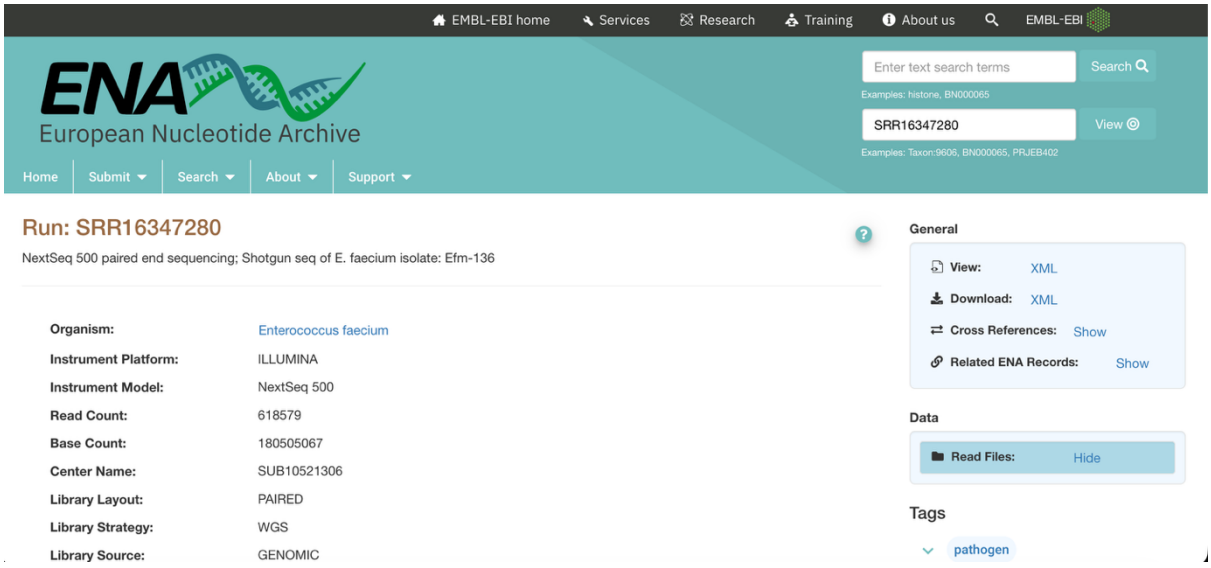
On your internet browser go to: https://www.ebi.ac.uk/ena/browser/home



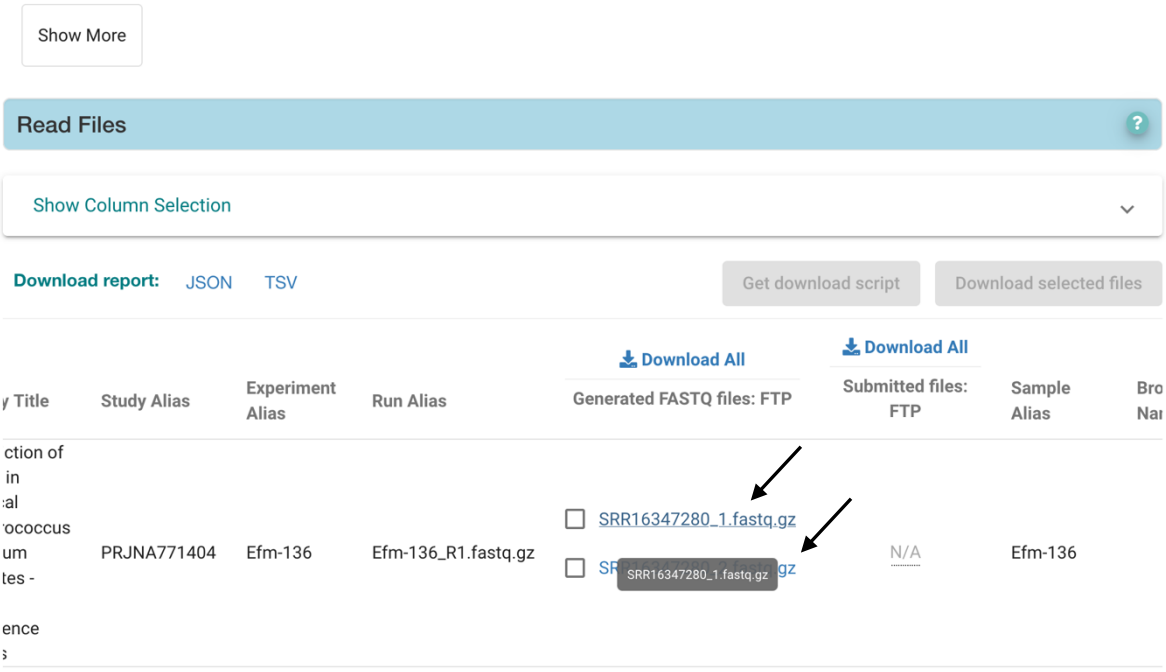

Type in a run accession into the search box as shown below and click on "View":




A new page will show all the sequencing information related to that run accession.




Scroll down and right to visualise information on the read files. Now click on each of the fastq files (_1.fastq.gz and _2.fastq.gz) under “Generated FASTQ files: FTP” to download both files.



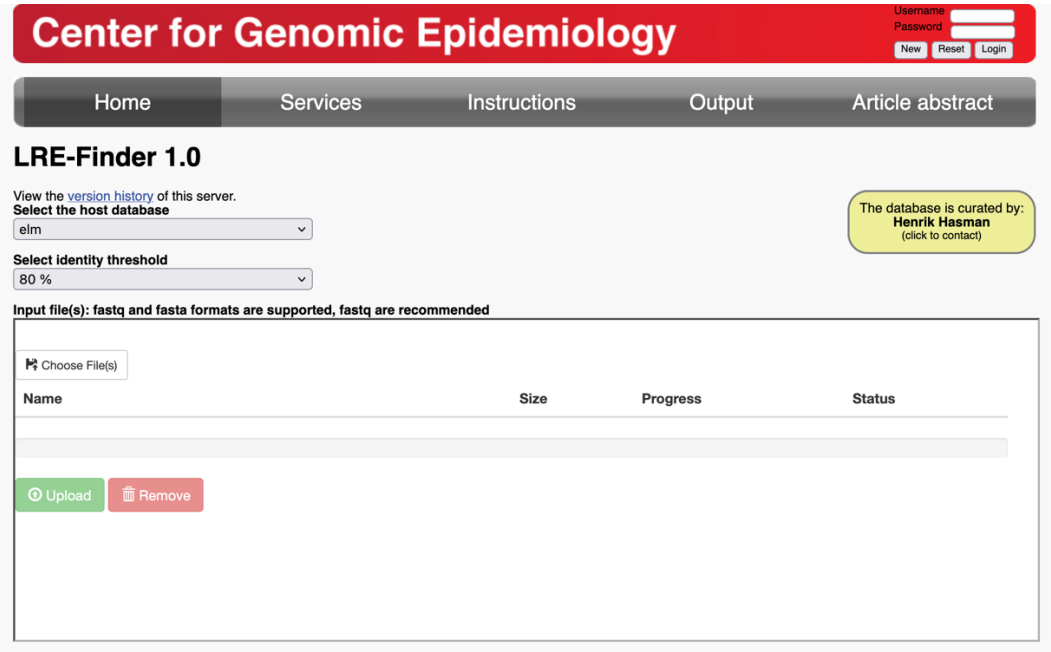

Repeat these steps to download the fastq files of all four *E. faecium* strains in Table 1.

### 1.2 Accessing genome assemblies

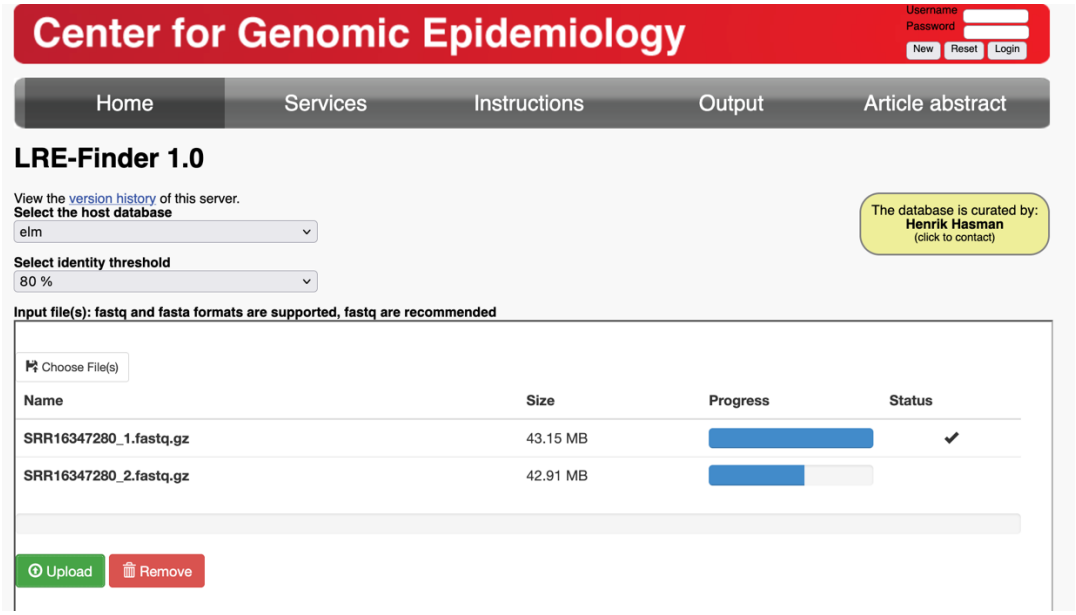
We have predownloaded the genome assemblies of all four strains from AllTheBacteria, which are available on the GitHub directory of this practical: https://github.com/francesccoll/teaching/tree/main/upv_practical/e_faecium

### 1.3 Running LRE-Finder from raw Illumina sequence reads

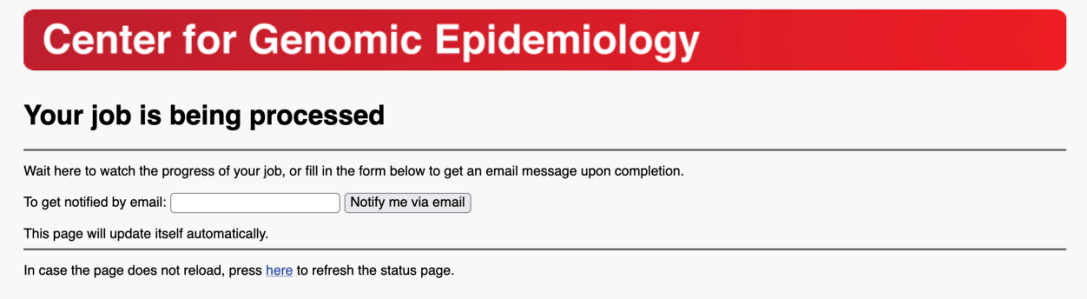
Next, we will used a specialized tool called LRE-Finder, a tool developed to detect genetic markers of linezolid resistance in enterococci from whole-genome sequences. This tool accepts both raw Illumina reads and genome assemblies as input, and has both a command-line and online version. Go to: https://cge.food.dtu.dk/services/LRE-finder/




Click on "Choose File(s)", browse to the directory where fastq.gz files were downloaded and select both the forward (_1.fastq.gz) and reverse (_2.fastq.gz) reads.

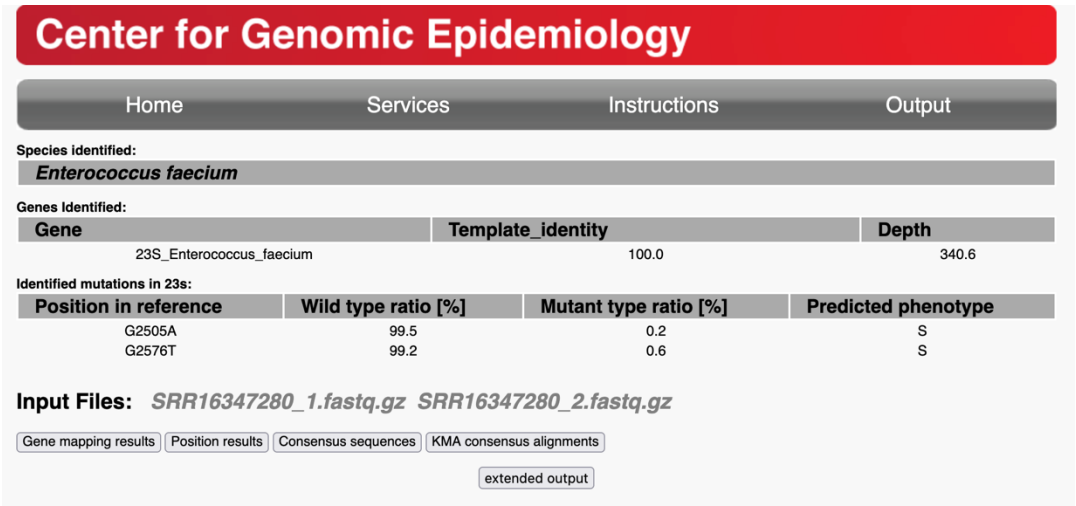


Once both fastq.gz files have been uploaded, a new window like the one below will appear:




Keep the window open while the sequence data is being processed.

The results page of LRE-Finder should look like the one below:




Take your time to understand what information each results field conveys, particularly those related to “Identified mutations in 23S”.

If fastq files take a long time to download or to be processed by LRE-Finder, you can access the results of previously uploaded data using the links in Table 2.

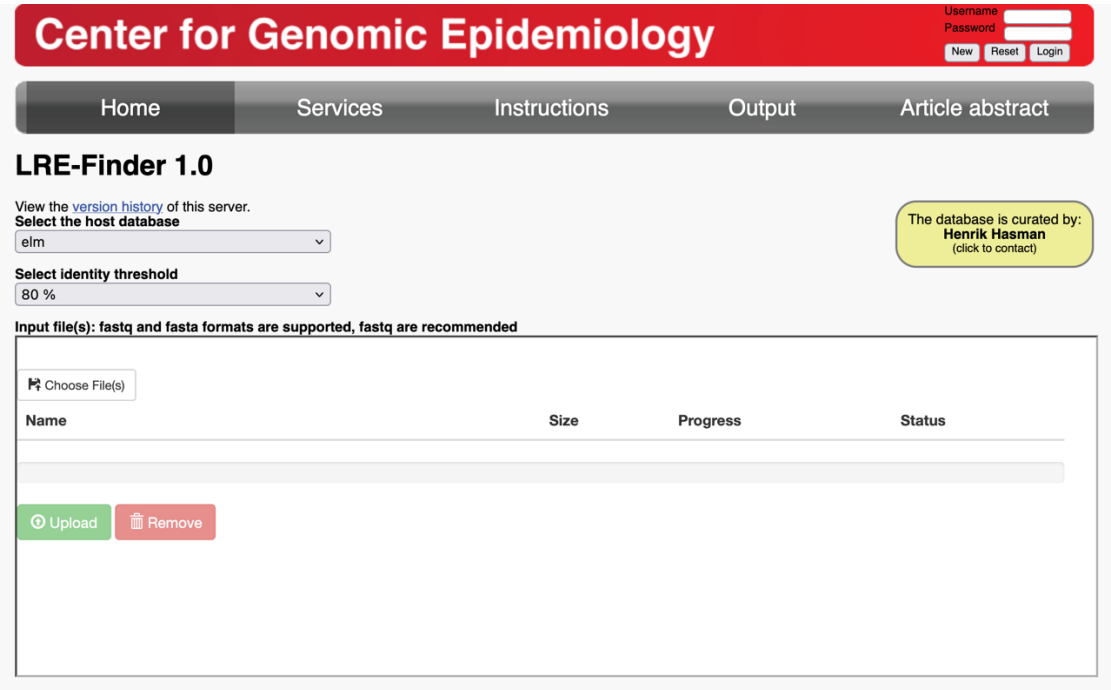
**Table 2**. LRE-Finder results from *E. faecium* raw Illumina sequence reads.

| Run accession | LRE-Finder URL |
|---|---|
| SRR16347280 | https://cge.food.dtu.dk//cgi-bin/webface.fcgi?jobid=6ACBBEF70000482246915A30 |
| SRR16347263 | https://cge.food.dtu.dk//cgi-bin/webface.fcgi?jobid=6ACBC0B3000048B56FCBF470 |
| SRR16347013 | https://cge.food.dtu.dk//cgi-bin/webface.fcgi?jobid=6ACBC1B4000048FFFF2200B2 |
| SRR16346985 | https://cge.food.dtu.dk//cgi-bin/webface.fcgi?jobid=6ACBC1BF000049134202C5A7 |

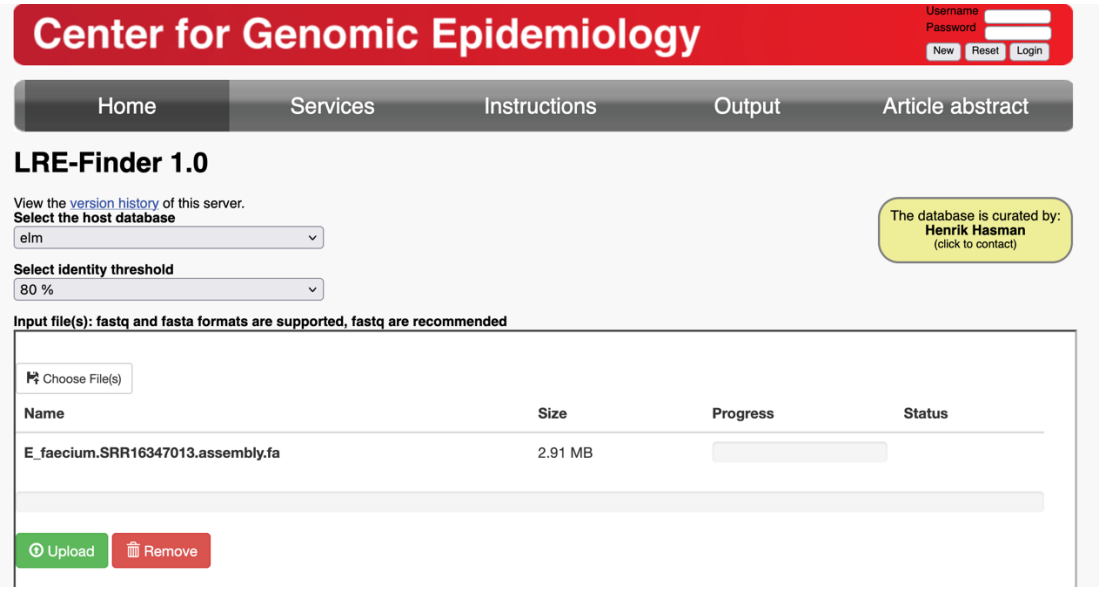
Take note of the “Identified mutations in 23S”, “Wild type ration [%]”, “Mutant type ratio [%]” and “Predicted phenotype” in each of the four genomes analysed.

### 1.4 Running LRE-Finder from Illumina assemblies

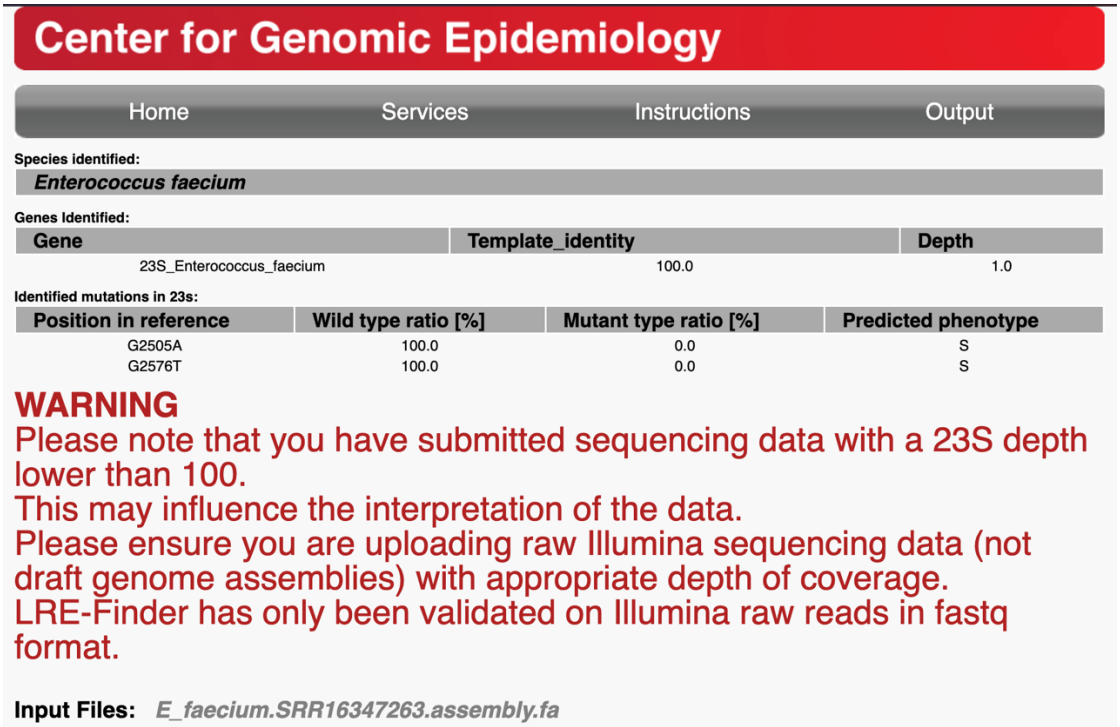
Open up a new tab on your Internet browser and go to:
https://cge.food.dtu.dk/services/LRE-finder/




Click on "Choose File(s)", browse to the directory where you downloaded the Illumina assemblies from GitHub. Select only one assembly file at the time, as shown in the screenshot below, and click on "Upload". Repeat these steps for each of the four assemblies on different tabs.




The results page of LRE-Finder should look like the one below:



Center for Genomic Epidemiology

Home | Services | Instructions | Output

Species identified:

*Enterococcus faecium*

Genes Identified:

| Gene | Template_identity | Depth |
|---|---|---|
| 23S_Enterococcus_faecium | 100.0 | 1.0 |

Identified mutations in 23s:

| Position in reference | Wild type ratio [%] | Mutant type ratio [%] | Predicted phenotype |
|---|---|---|---|
| G2505A | 100.0 | 0.0 | S |
| G2576T | 100.0 | 0.0 | S |

**WARNING**

Please note that you have submitted sequencing data with a 23S depth lower than 100.
This may influence the interpretation of the data.
Please ensure you are uploading raw Illumina sequencing data (not draft genome assemblies) with appropriate depth of coverage.
LRE-Finder has only been validated on Illumina raw reads in fastq format.

**Input Files:** *E_faecium.SRR16347263.assembly.fa*

If assemblies take a long time to be processed by LRE-Finder, you can you can access the results of previously uploaded assemblies using the links below:

Table 3. LRE-Finder results from *E. faecium* genome assemblies.

| Assembly | LRE-Finder URL |
|---|---|
| E_faecium.SRR16346985.assembly.fa | https://cge.food.dtu.dk//cgi-bin/webface.fcgi?jobid=6ACBC40F00004984E91BDD55 |
| E_faecium.SRR16347263.assembly.fa | https://cge.food.dtu.dk//cgi-bin/webface.fcgi?jobid=6ACBC3FD00004965718E9461 |
| E_faecium.SRR16347013.assembly.fa | https://cge.food.dtu.dk//cgi-bin/webface.fcgi?jobid=6ACBC39400004947CF34565F |
| E_faecium.SRR16347280.assembly.fa | https://cge.food.dtu.dk//cgi-bin/webface.fcgi?jobid=6ACBC40700004974B3D64519 |

**? Questions**

Can phenotypic resistance to linezolid (i.e. that presented in “Linezolid phenotypic resistance” in Table 1) be explained by known genetic markers of resistance?

Are there any discrepancies in the linezolid resistance mutations found and in the predicted phenotypes between for the same strain when using raw sequencing reads or their genome assemblies? if so, what could explain these discrepancies?

Are discrepancies in predicted phenotypes present in all four strains? What could explain the presence or absence of genotype-phenotype discrepancies between strains when using Illumina genome assemblies?

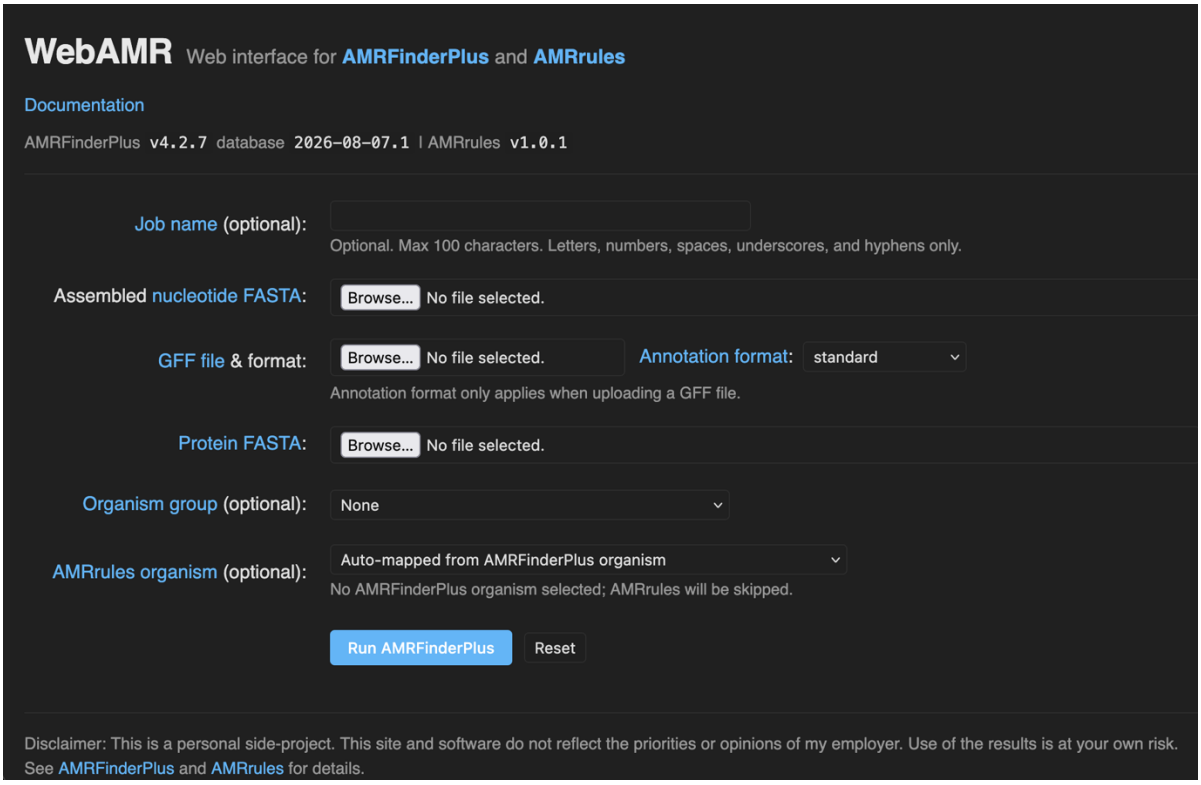
## Case study 2: Effect of duplicated and truncated AMR genes on genotypic AMR determination

Tetracycline resistance in *Enterococcus faecium* and other Gram-positive bacteria is primarily mediated by ribosomal protection proteins encoded by *tet(M)* and efflux pumps encoded by *tet(L)*. *tet(M)* is among the most prevalent tetracycline resistance genes in this species and is often carried on mobile genetic elements (MGEs). The authors of a recent study identified many false negatives when predicting tetracycline resistance from Illumina genome assemblies. They noticed that many of these **false negatives** (i.e. tetracycline resistant strains by phenotypic antibiotic susceptibility testing lacking *tet* genes in their genome) actually carried partially detected *tet(M)* genes. In this exercise, we will interrogate Illumina and hybrid assemblies to investigate the source of tetracycline genotype-phenotype discrepancies.

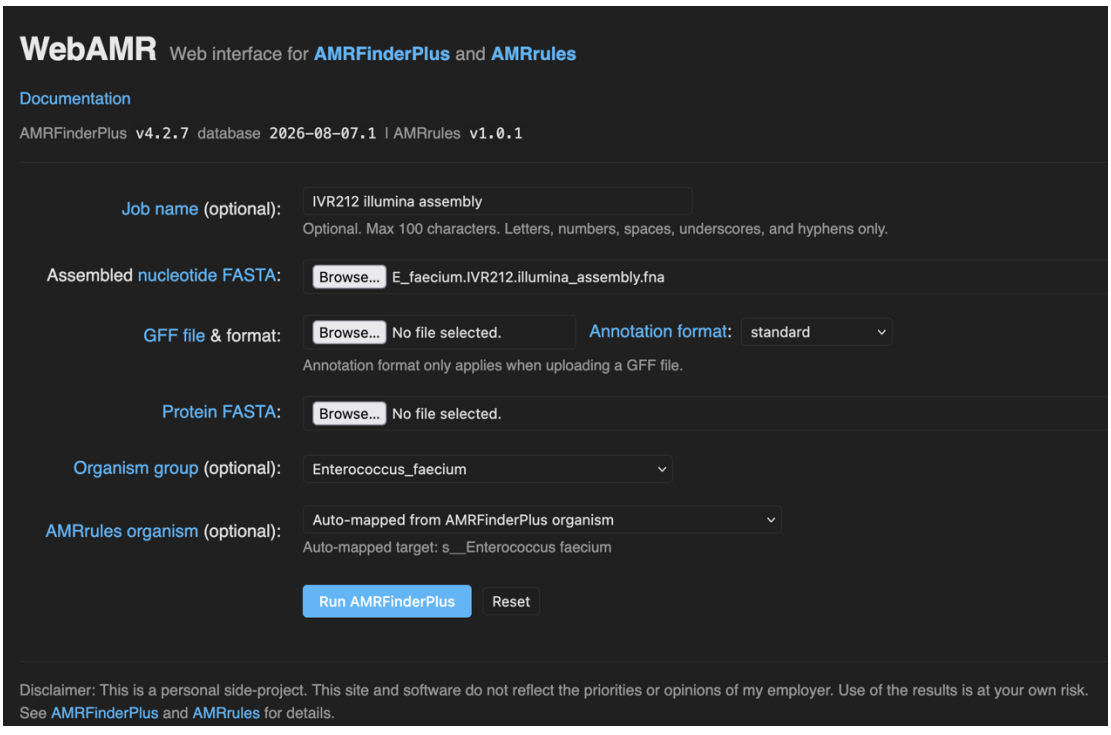
A **hybrid assembly** is a genome assembly method that combines sequencing data from two different technologies, typically short reads (e.g. Illumina, which are highly accurate) and long reads (e.g. ONT, which cover longer regions of DNA). By combining their strengths, it can produce a more complete and accurate genome assembly than using either type of data alone. A hybrid assembly usually produces a **closed genome**, meaning that the genome is assembled into a single, continuous DNA sequence for each chromosome or replicon, without gaps or unresolved regions. In this exercise, we will analyse the resistome of three *E. faecium* strains, for which both Illumina-only and hybrid (Illumina + ONT) genome assemblies are available. Download the Illumina assemblies (files ending with ".illumina_assembly.fna") and hybrid assemblies (files ending in ".hybrid_assembly.fna") of the three strains (IVR212, VREN0576 and VREN1631) available on the GitHub directory of this practical:
https://github.com/francesccoll/teaching/tree/main/upv_practical/e_faecium

In this case we will use AMRFinderPlus, a bioinformatics tool from the NCBI developed to detect antimicrobial resistance (AMR) genes and resistance-associated mutations in bacterial genomes or protein sequences. AMRFinderPlus is originally developed as a command-line tool that only takes assemblies as input files, it does not support taking raw reads as input. We will run and interpret AMRFinderPlus reports using a web interface: https://amr.arjunp.net/.

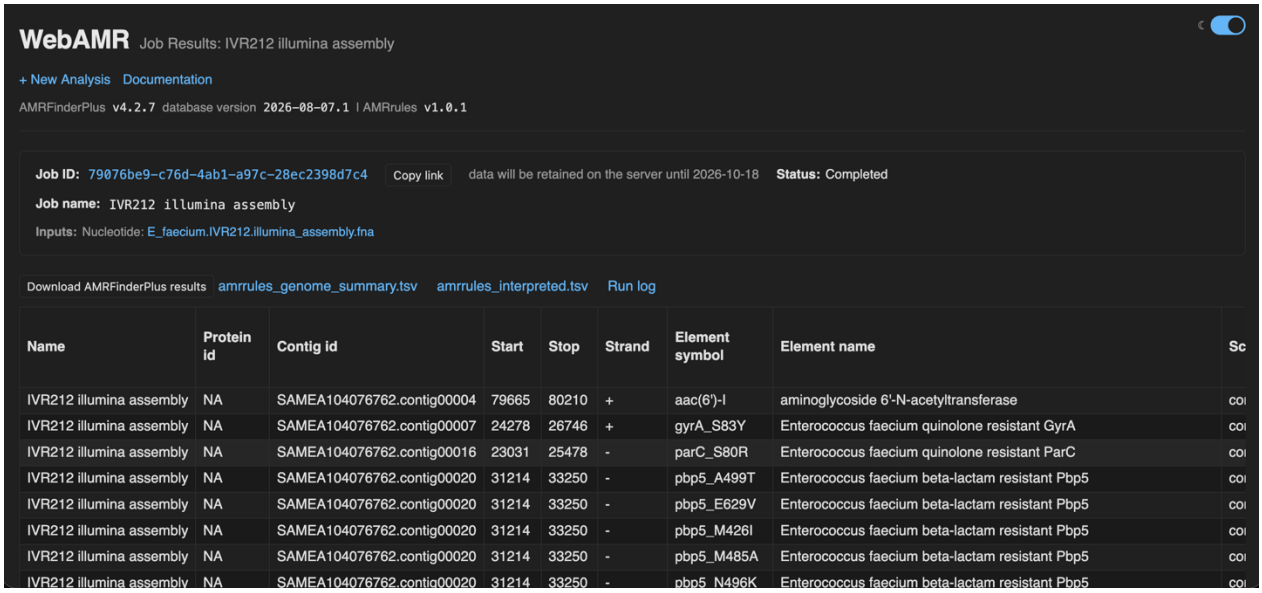
Go to https://amr.arjunp.net/ using your internet browser. You should be able to see a web page like the one above.




Click on the "Browse" button next to "Assembled nucleotide FASTA" to browse and select for one of the assemblies to be analysed, choose a "Job name" that reflects the strain and type of assembly you are analysing (e.g. "IVR212 illumina assembly"), select "Enterococcus_faecium" under "Organism group (optional)", and finally click on the blue button "Run AMRFinderPlus". See the screenshot below as an example.



After a few minutes a new window with the AMRFinderPlus results will appear. Remember to copy and save the resulting URL (starting with "https://amr.arjunp.net/results/" followed by the Job Id) to retrieve the results of your submitted assembly any time.




| Name | Protein id | Contig id | Start | Stop | Strand | Element symbol | Element name | Sc |
|---|---|---|---|---|---|---|---|---|
| IVR212 illumina assembly | NA | SAMEA104076762.contig00004 | 79665 | 80210 | + | aac(6')-Ii | aminoglycoside 6'-N-acetyltransferase | co |
| IVR212 illumina assembly | NA | SAMEA104076762.contig00007 | 24278 | 26746 | + | gyrA_S83Y | Enterococcus faecium quinolone resistant GyrA | co |
| IVR212 illumina assembly | NA | SAMEA104076762.contig00016 | 23031 | 25478 | - | parC_S80R | Enterococcus faecium quinolone resistant ParC | co |
| IVR212 illumina assembly | NA | SAMEA104076762.contig00020 | 31214 | 33250 | - | pbp5_A499T | Enterococcus faecium beta-lactam resistant Pbp5 | co |
| IVR212 illumina assembly | NA | SAMEA104076762.contig00020 | 31214 | 33250 | - | pbp5_E629V | Enterococcus faecium beta-lactam resistant Pbp5 | co |
| IVR212 illumina assembly | NA | SAMEA104076762.contig00020 | 31214 | 33250 | - | pbp5_M426I | Enterococcus faecium beta-lactam resistant Pbp5 | co |
| IVR212 illumina assembly | NA | SAMEA104076762.contig00020 | 31214 | 33250 | - | pbp5_M485A | Enterococcus faecium beta-lactam resistant Pbp5 | co |
| IVR212 illumina assembly | NA | SAMEA104076762.contig00020 | 31214 | 33250 | - | pbp5_N496K | Enterococcus faecium beta-lactam resistant Pbp5 | co |

You should now have obtained a results page like the screenshot above. Take you time to browse and explore the resulting table.

You can find a description of all fields in the AMRFinderPlus output report here: https://github.com/ncbi/amr/wiki/Running-AMRFinderPlus#fields

Now repeat the same step for the three Illumina assemblies (files ending with ".illumina_assembly.fna") and three hybrid assemblies (files ending in ".hybrid_assembly.fna") of all strains (IVR212, VREN0576 and VREN1631). The table below contains the results of previously submitted jobs for each of these six assemblies.

Table 4. AMRFinderPlus results from *E. faecium* Illumina and hybrid assemblies.

| Strain | Assembly | AMRFinderPlus results URL |
|---|---|---|
| IVR212 | Illumina | https://amr.arjunp.net/results/2e0ae45f-2f2a-4042-831d-460403841c7a |
| | Hybrid | https://amr.arjunp.net/results/bbfafddf-6da6-4e89-81f0-3159590dec13 |
| VREN0576 | Illumina | https://amr.arjunp.net/results/75c40351-ed5b-4112-9b70-32d4484ef4da |
| | Hybrid | https://amr.arjunp.net/results/a8c62459-173d-4476-bd96-a1863415d908 |
| VREN1631 | Illumina | https://amr.arjunp.net/results/81779c04-f6bd-40e8-ae31-fe64a38697b0 |
| | Hybrid | https://amr.arjunp.net/results/14223420-1d67-43b6-b98b-d26e04282c02 |

**? Questions**

Remember that all three strains analysed here in this exercise are phenotypically resistant to tetracycline, therefore we should expect a genetic marker that explains that resistance, most often *tet* genes.

Has AMRFinderPlus identified any *tet* gene in the hybrid assemblies? How many copies and in what coverage? (here consider the fields "% Coverage of reference" and "% Identity to reference")

Based on the results of AMRFinderPlus from Illumina and hybrid assemblies, which results would you trust more? what could explain the lack of *tet(M)* detection in the Illumina assemblies?

## Case study 3. Resolving AMR phenotype-genotype discrepancies from *Salmonella enterica* genomes

In this exercise we will use Pathogenwatch to detect AMR from whole genome sequences of *Salmonella enterica* with a focus on cases of AMR genotype-phenotype discrepancies described in this study.

The tables below include the phenotypic antibiogram of four *Salmonella enterica* strains.

Table 5. Phenotypic resistance of *Salmonella enterica* strains.

| Strain ID | Ampicillin | Piperacillin | Amoxicillin/clavulanate | Chloramphenicol | Colistin | Fosfomycin |
|---|---|---|---|---|---|---|
| B27 | >8 (R) | >16 (R) | >8/4 (R) | >8 (R) | >2 (R) | >32 (R) |
| WD17 | >8 (R) | 16 (I) | ≤2/1 (S) | ≤8 (S) | ≤2 (S) | ≤32 (S) |
| WT5 | >8 (R) | >16 (R) | >8/4 (R) | >8 (R) | ≤2 (S) | ≤32 (S) |
| WW28B | >8 (R) | >16 (R) | 8/4 (S) | >8 (R) | ≤2 (S) | >32 (R) |

| Strain ID | Trimethoprim/sulfamethoxazole | Amikacin | Gentamicin | Tobramycin | Tetracycline |
|---|---|---|---|---|---|
| B27 | >4/76 (R) | ≤8 (S) | >4 (R) | >4 (R) | >8 (R) |
| WD17 | ≤2/38 (S) | ≤8 (S) | ≤2 (S) | ≤2 (S) | ≤4 (S) |
| WT5 | >4/76 (R) | ≤8 (S) | ≤2 (S) | ≤2 (S) | >8 (R) |
| WW28B | >4/76 (R) | ≤8 (S) | ≤2 (S) | ≤2 (S) | >8 (R) |

And the table below reported cases of AMR genotype-phenotype discrepancies we will investigate:

Table 6. AMR genotype-phenotype discrepancies for *Salmonella enterica* strains.

| Strain ID | Nature of Error | Genotype-Phenotype Disagreement |
|---|---|---|
| B27 | False negative | Amoxicillin/clavulanate resistance without any resistance determinant |
| B27 | False negative | Colistin resistance without any resistance determinant |
| B27 | False positive | Amikacin susceptibility despite presence of *aac(6')-Iaa* gene |
| WD17 | False negative | Ampicillin resistance without any resistance determinant |
| WD17 | False positive | Amikacin and tobramycin susceptibility despite presence of *aac(6')-Iaa* gene |
| WT5 | False negative | Amoxicillin/clavulanate resistance without any resistance determinant |
| WT5 | False positive | Amikacin and tobramycin susceptibility despite presence of *aac(6')-Iaa* gene |
| WW28B | False negative | Fosfomycin resistance without any resistance determinant |
| WW28B | False positive | Amikacin and tobramycin susceptibility despite presence of *aac(6')-Iaa* gene |

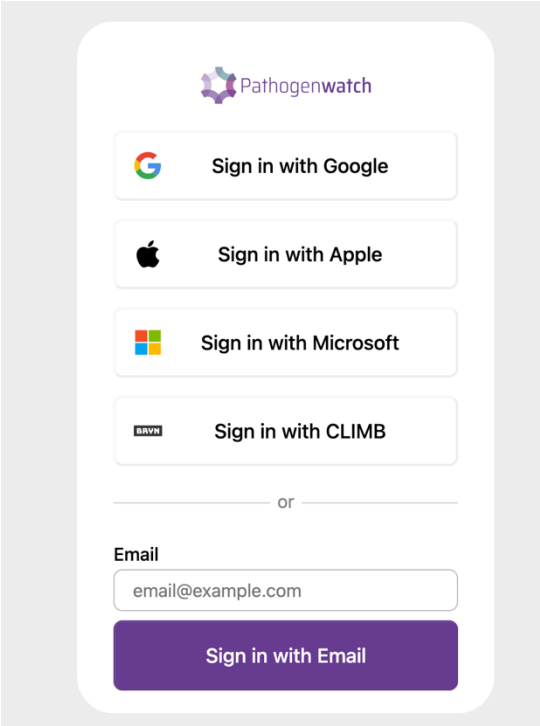
Download the assemblies of the four *Salmonella enterica* strains from GitHub: https://github.com/francesccoll/teaching/tree/main/upv_practical/salmonella

Use your browser to navigate to https://pathogen.watch/.

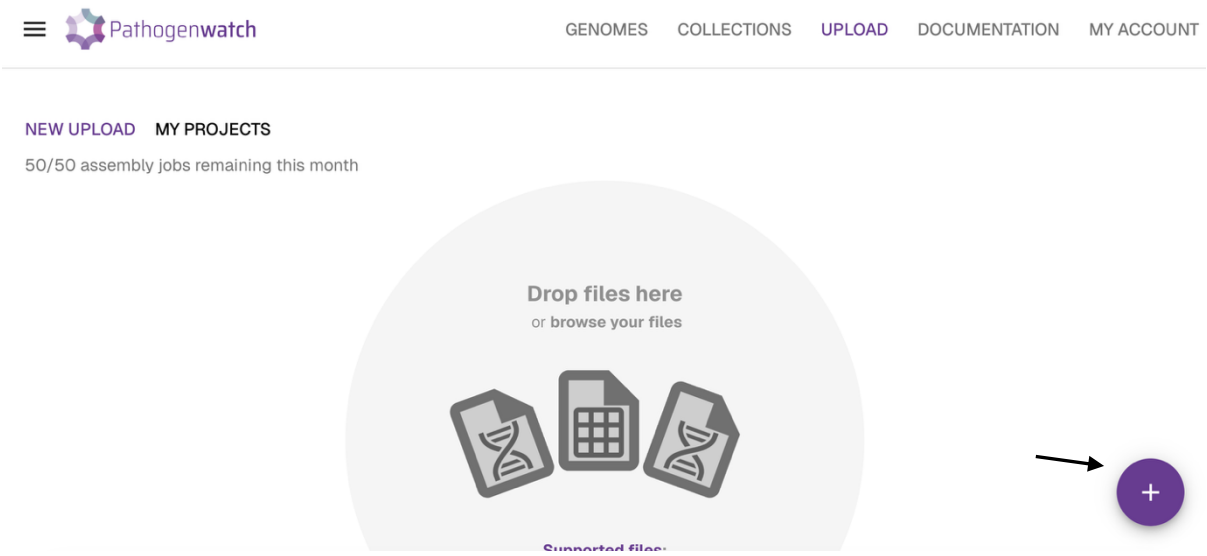
At the top-right corner of the page click on “UPLOAD” and click on “SIGN IN”.



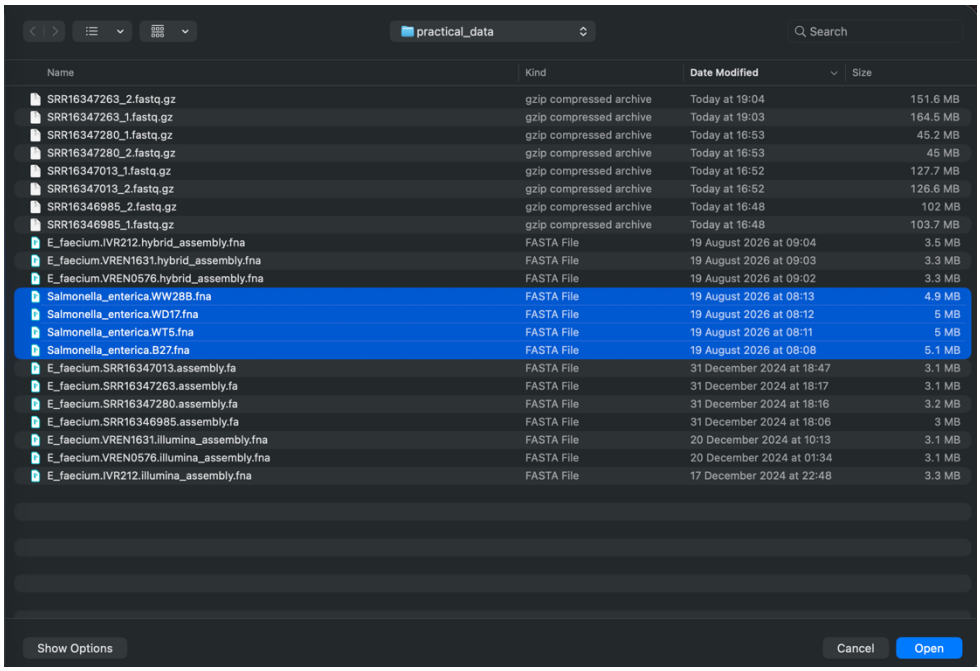

Choose your preferred option to signed in.



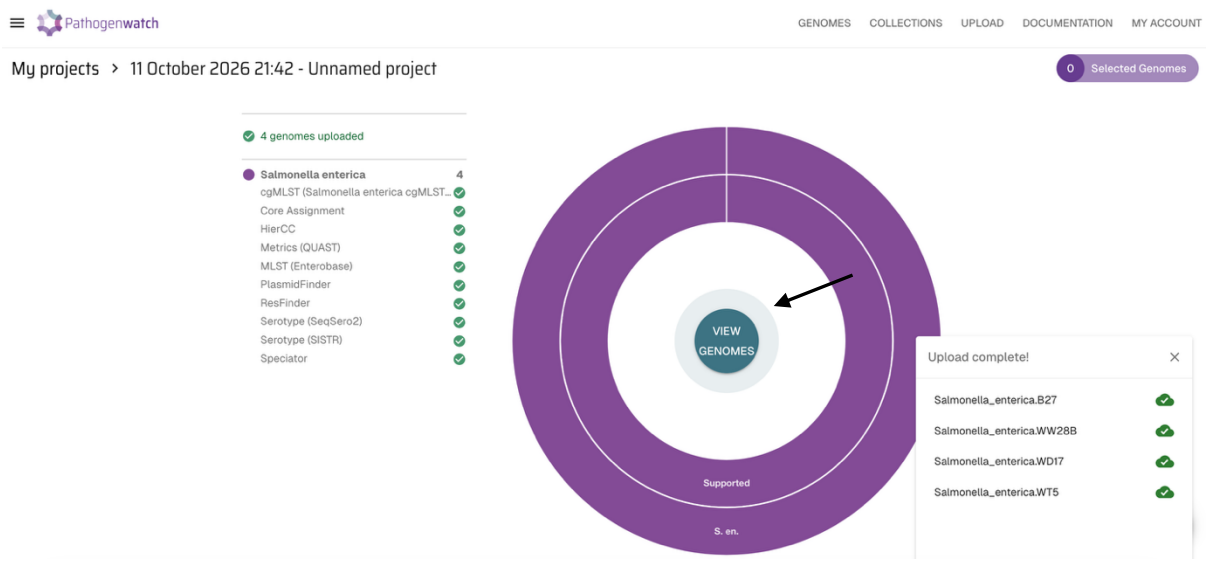

After signing in, click on the "+" button at the bottom-right corner of the page.



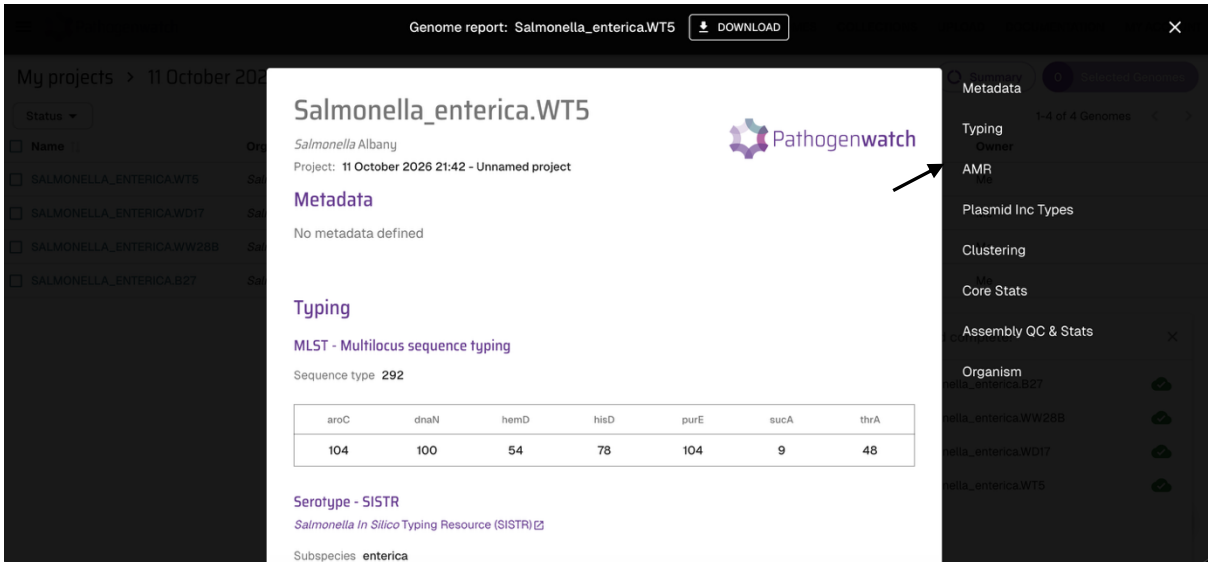
Select all four assemblies as shown below.




The genome assemblies will be automatically analysed by Pathogenwatch. Click on “VIEW GENOMES” after the processing has finished.




Click on each genome “Name” to visualise the “Genome report” of each strain at a time. Familiarise yourself with the different sections of the report and navigate to the “AMR” section.




Genome reports can be downloaded as PDFs. To facilitate the comparison between all four reports, open up each report in a different tab on your browser or download all reports as PDFs. Take a look at the reported phenotypic resistance of strains in Table 5 and the predicted genotypic resistance on the Pathogenwatch reports to answer the questions below.

### ? Questions

All four *Salmonella* strains are resistant to ampicillin according to phenotypic testing. Can you identify which resistance determinant(s) are responsible for phenotypic resistance to ampicillin in each strain? Do all four strains carry at least one resistance determinant that could explain ampicillin resistance?

Three *Salmonella* strains (B27, WT5, and WW28B) are resistant to trimethoprim/sulfamethoxazole. Can you identify the AMR genes responsible for resistance to this antibiotic combination?

Two *Salmonella* strains (B27 and WW28B) are resistant to fosfomycin. Can you identify the AMR gene(s) responsible for resistance to this antibiotic?

For false negatives, i.e. phenotypically resistant strains without any resistance determinant detected by PathogenWatch, could you hypothesise what biological cause(s) may be responsible for such discrepancies?